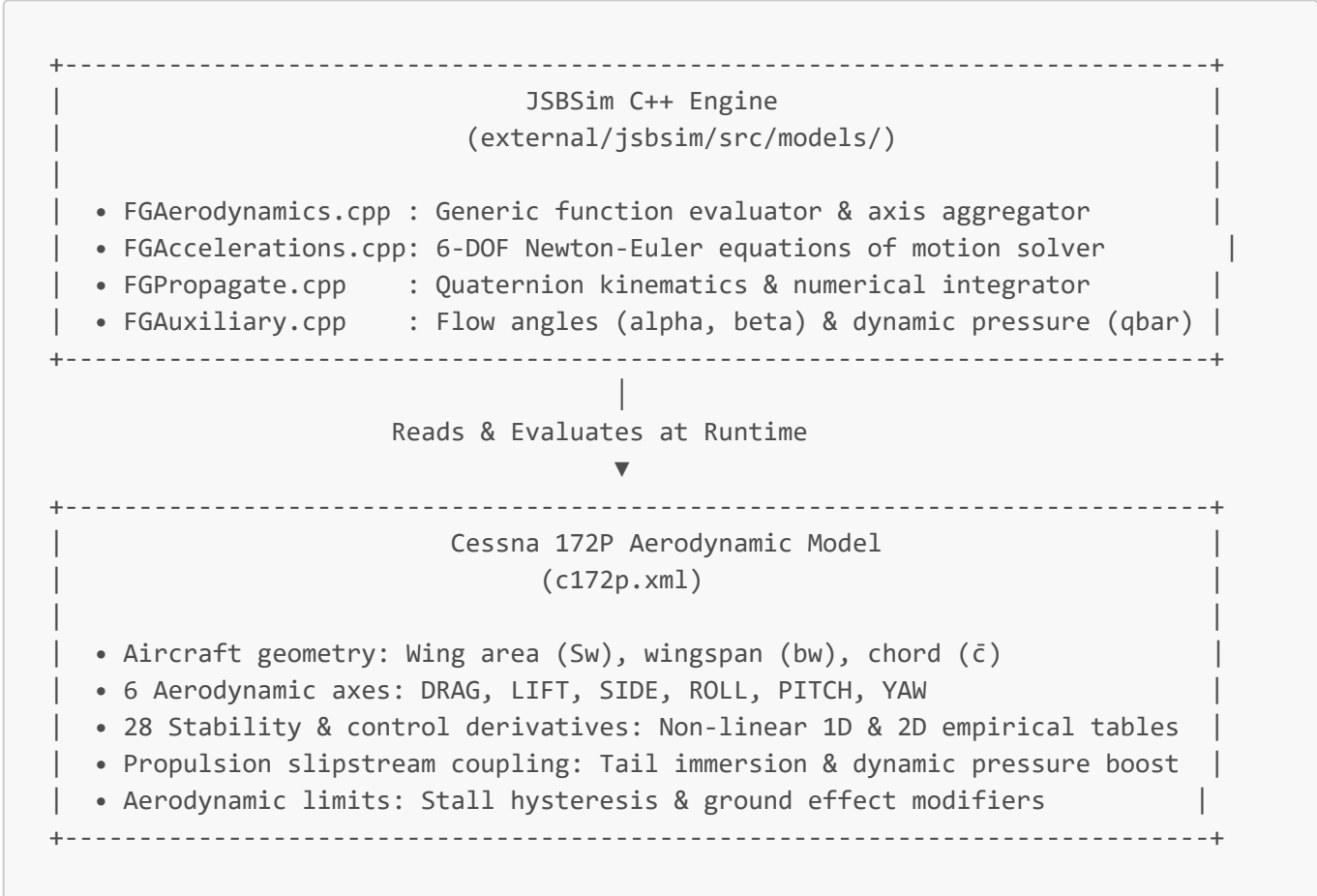
Cessna 172 Skyhawk: Aerodynamic Coefficients, Stability Derivatives & Mathematical Formulations

Platform: Cessna 172P Skyhawk (`c172p`)
Engine: JSBSim 6-DOF Flight Dynamics Model
Source XML Configuration: c172p.xml
C++ Physics Implementation: FGAerodynamics.cpp

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1. Architectural Division: JSBSim C++ Engine vs. Cessna XML Database

A fundamental design principle of JSBSim is the complete decoupling between the **generic physics engine** and the **aircraft aerodynamic database**:



- **Inside JSBSim C++**: Contains zero hardcoded numbers for the Cessna 172. It defines 6 abstract axes (`LIFT`, `DRAG`, `SIDE`, `ROLL`, `PITCH`, `YAW`), evaluates arbitrary XML mathematical trees, transforms forces across coordinate frames (Wind → Stability → Body), and transfers moments to the Center of Gravity (CG).
- **Inside Cessna XML (`c172p.xml`)**: Contains all numerical constants, stability derivatives, empirical lookup tables, control sensitivities, and flap increments specific to the Cessna 172P.

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2. Reference Metrics & Geometric Parameters

All aerodynamic forces and moments scale with the reference wing geometry defined in the `<metrics>` block (lines 26–48):

Classical Geometry Table

Parameter	Symbol	JSBSim Property	Value	Units	Description
Wing Reference Area	Sw	`metrics/Sw-sqft`	174.00	sq ft	Total planform wing area
Wingspan	bw	`metrics/bw-ft`	35.80	ft	Tip-to-tip wing span
Mean Aerodynamic Chord	c̄	`metrics/cbarw-ft`	4.90	ft	Reference longitudinal chord
Wing Aspect Ratio	AR	Computed	7.37	-	AR = bw^2 / Sw
Wing Incidence Angle	iw	`metrics/iw-deg`	1.50	deg	Wing mounting deck angle
Aerodynamic Ref Point (X)	xRP	`metrics/aero-rp-x-in`	43.80	in	Longitudinal aero center from datum
Aerodynamic Ref Point (Y)	yRP	`metrics/aero-rp-y-in`	0.00	in	Lateral symmetry centerline
Aerodynamic Ref Point (Z)	zRP	`metrics/aero-rp-z-in`	58.70	in	Vertical aero center from datum

3. Kinematic Normalization & Flow Parameters

Mathematical Symbolic Equations:

```
1. Free-Stream Dynamic Pressure:
   q̄ = ½ · ρ · Vt²

2. Propeller Slipstream Induced Velocity:
   Vinduced = uaero + 2 · Vprop_induced

3. Slipstream Dynamic Pressure at Tail (Propwash):
   q̄ind = ½ · ρ · (Vinduced)²

4. Non-Dimensional Angular Body Rates:
   p̂ = (p · bw) / (2 · Vt)       [Roll rate damping ratio]
   q̂ = (q · c̄)  / (2 · Vt)       [Pitch rate damping ratio]
   r̂ = (r · bw) / (2 · Vt)       [Yaw rate damping ratio]

5. Non-Dimensional Alpha Rate:
   α̂̇ = (α̇ · c̄)  / (2 · Vt)       [Unsteady downwash lag ratio]
```

JSBSim Implementation Properties:

```
qbar         = aero/qbar-psf                         (psf)
qbar_induced = aero/function/qbar-induced-psf        (psf)
p_hat        = aero/bi2vel * velocities/p-aero-rad_sec
q_hat        = aero/ci2vel * velocities/q-aero-rad_sec
r_hat        = aero/bi2vel * velocities/r-aero-rad_sec
alphadot_hat = aero/ci2vel * aero/alphadot-rad_sec
```

Aerodynamic Boundaries (lines 401–409):

```
αmin = -0.087 rad  (-4.98°)
αmax = +0.280 rad  (+16.04°, critical stall threshold αcrit)

αhyst_min = +0.090 rad  (+5.16°)
αhyst_max = +0.360 rad  (+20.63°)
```

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4. Ground Effect Modifiers

Ground effect modifies lift and induced drag as a function of the normalized wheel clearance above the runway surface:

```
Normalized Height Ratio:
ĥ = hgear / c̄ = aero/h_b-mac-ft
```

Mathematical Scaling Equations:

```
CDinduced(in ground effect) = kCDge(ĥ) · CDinduced(free stream)
CLcirc(in ground effect)    = kCLge(ĥ) · CLcirc(free stream)
```

Ground Effect Factor Table: kCDge and kCLge (lines 411–454)

Height Ratio ĥ = h / c̄	kCDge (Drag Multiplier)	kCLge (Lift Multiplier)	Physical Flight State
0.0000	**0.4800**	**1.2030**	Touchdown: -52% induced drag, +20.3% lift cushion
0.1000	0.5150	1.1270	Wheel height ~0.5 ft: strong ground cushion
0.1500	0.6290	1.0900	Normal landing flare roundout
0.2000	0.7090	1.0730	Takeoff rotation unstick height

Height Ratio ĥ = h / c̄	kCDge (Drag Multiplier)	kCLge (Lift Multiplier)	Physical Flight State
0.3000	0.8150	1.0460	Initial climb-out / obstacle screen
0.4000	0.8820	1.0550	Transition height
0.5000	0.9280	1.0190	Half-chord clearance
0.6000	0.9620	1.0130	Weakening ground effect
0.7000	0.9880	1.0080	Residual ground effect
0.8000	1.0000	1.0060	Near free-stream
0.9000	1.0000	1.0030	Free-stream boundary
1.0000	1.0000	1.0020	Out of ground effect (~36 ft AGL)
1.1000+	1.0000	1.0000	Pure free-stream aerodynamic flow

5. Axis: DRAG (Wind Axis)

Mathematical Symbolic Equation:

```
D = q̄ · Sw · CD

Where the total drag coefficient CD is decomposed as:
CD = CD0 + kCDge(ĥ) · [ ΔCDflap(δf) + CDwbh(α, δf) ] + CDδe · |δe| + CDβ · |β|
```

Stability & Control Derivatives:

```
Zero-Lift Parasite Drag:       CD0   = 0.0270
Elevator Deflection Drag:      CDδe  = ∂CD / ∂|δe| = 0.0600 per rad  (0.00105 /
deg)
Sideslip Parasite Drag:        CDβ   = ∂CD / ∂|β|  = 0.1700 per rad  (0.00297 /
deg)
```

Components & Tables:

1. **Zero-Lift Parasite Drag (CD0)**:
   - Symbol: `CD0 = 0.0270`
   - JSBSim Property: `aero/coefficient/CDo`
   - Formula: `Drag0 = q̄ · Sw · 0.0270`
2. **Flap Profile Drag Increment (ΔCDflap)**:
   - JSBSim Property: `aero/coefficient/CDDf`

- Table:

Flap Position δf	ΔCDflap	Physical Meaning
0.0°	0.0000	Clean wing baseline
10.0°	0.0070	Takeoff flap profile drag
20.0°	0.0120	Approach flap profile drag
30.0°	0.0180	Full landing flap profile drag

3. **Induced Drag vs. Angle of Attack & Flaps (CDwbh)**:

- JSBSim Property: `aero/coefficient/CDwbh`
- Full 2D Empirical Table:

α (rad)	α (deg)	Flaps 0°	Flaps 10°	Flaps 20°	Flaps 30°
-0.0873	-5.00°	0.0041	0.0000	0.0005	0.0014
-0.0698	-4.00°	0.0013	0.0004	0.0025	0.0041
-0.0524	-3.00°	0.0001	0.0023	0.0059	0.0084
-0.0349	-2.00°	0.0003	0.0057	0.0108	0.0141
-0.0175	-1.00°	0.0020	0.0105	0.0172	0.0212
0.0000	0.00°	0.0052	0.0168	0.0251	0.0299
0.0175	1.00°	0.0099	0.0248	0.0346	0.0402
0.0349	2.00°	0.0162	0.0342	0.0457	0.0521
0.0524	3.00°	0.0240	0.0452	0.0583	0.0655
0.0698	4.00°	0.0334	0.0577	0.0724	0.0804
0.0873	5.00°	0.0442	0.0718	0.0881	0.0968
0.1047	6.00°	0.0566	0.0874	0.1053	0.1148
0.1222	7.00°	0.0706	0.1045	0.1240	0.1343
0.1396	8.00°	0.0860	0.1232	0.1442	0.1554
0.1571	9.00°	0.0962	0.1353	0.1573	0.1690
0.1745	10.00°	0.1069	0.1479	0.1708	0.1830
0.1920	11.00°	0.1180	0.1610	0.1849	0.1975
0.2094	12.00°	0.1298	0.1746	0.1995	0.2126
0.2269	13.00°	0.1424	0.1892	0.2151	0.2286
0.2443	14.00°	0.1565	0.2054	0.2323	0.2464
0.2618	15.00°	0.1727	0.2240	0.2521	0.2667

α (rad)	α (deg)	Flaps 0°	Flaps 10°	Flaps 20°	Flaps 30°
0.2793	**16.00°**	**0.1782**	**0.2302**	**0.2587**	**0.2735**
0.2967	17.00°	0.1716	0.2227	0.2507	0.2653
0.3142	18.00°	0.1618	0.2115	0.2388	0.2531
0.3316	19.00°	0.1475	0.1951	0.2214	0.2351
0.3491	20.00°	0.1097	0.1512	0.1744	0.1866

4. **Elevator Deflection Drag**:
    - JSBSim Property: `aero/coefficient/CDDe`
    - Formula: `ΔCD = 0.0600 · |δe|`
5. **Sideslip Drag Penalty**:
    - JSBSim Property: `aero/coefficient/CDbeta`
    - Formula: `ΔCD = 0.1700 · |β|`

---

6. Axis: LIFT (Wind Axis)

Mathematical Symbolic Equation:

```
L = q̄ · Sw · CL

Where the total lift coefficient CL is decomposed as:
CL = kCLge(ĥ) · [ CLwbh(α, hyst) + ΔCLflap(δf) ] + CLδe · δe + CLα̇ · α̂ + CLq · q̂
```

Stability & Control Derivatives:

```
Lift Curve Slope:              CLα   = ∂CL / ∂α   = 4.4700 per rad  (0.0780 / deg)
Elevator Lift Effectiveness:   CLδe  = ∂CL / ∂δe  = 0.4300 per rad  (0.0075 / deg)
Alpha-Rate Lift Lag:           CLα̇ = ∂CL / ∂α̂ = 1.7000
Pitch-Rate Lift Damping:       CLq   = ∂CL / ∂q̂   = 3.9000
Maximum Clean Lift:            CLmax = 1.4700 at αcrit = 16.04° (0.28 rad)
```

Components & Tables:

1. **Wing-Body Lift vs. Alpha & Stall Hysteresis (CLwbh)**:
    - JSBSim Property: `aero/coefficient/CLwbh`
    - Full 2D Table:

α (rad)	α (deg)	Attached Flow (hyst = 0.0)	Stalled Flow (hyst = 1.0)	Aerodynamic State

α (rad)	α (deg)	Attached Flow (hyst = 0.0)	Stalled Flow (hyst = 1.0)	Aerodynamic State
-0.0900	-5.16°	-0.2200	-0.2200	Zero-lift alpha α0L ≈ -2.8°
0.0000	0.00°	+0.2500	+0.2500	Camber lift at zero alpha
0.0900	5.16°	+0.7300	+0.7300	Normal cruise operating range
0.1000	5.73°	+0.8300	+0.7800	Hysteresis onset band
0.1200	6.88°	+0.9200	+0.7900	Attached linear lift slope
0.1400	8.02°	+1.0200	+0.8100	High angle of attack climb
0.1600	9.17°	+1.0800	+0.8200	Slow flight regime
0.1700	9.74°	+1.1300	+0.8300	Approaching buffet onset
0.1900	10.89°	+1.1900	+0.8500	Stall warning horn threshold
0.2100	12.03°	+1.2500	+0.8600	Moderate flow separation at trailing edge
0.2400	13.75°	+1.3500	+0.8800	Severe flow separation
0.2600	14.90°	+1.4400	+0.9000	Peak buffet
0.2800	**16.04°**	**1.4700 (CLmax)**	**0.9200 (Stall Break)**	**Critical Stall Angle of Attack**
0.3000	17.19°	+1.4300	+0.9500	Post-stall autorotation zone
0.3200	18.33°	+1.3800	+0.9900	Post-stall autorotation zone
0.3400	19.48°	+1.3000	+1.0500	Deep stall / developed spin
0.3600	20.63°	+1.1500	+1.1500	Complete upper-surface flow separation

2. **Flap Lift Increment (ΔCLflap)**:
    - JSBSim Property: `aero/coefficient/CLDf`
    - Table:

Flap Position δf	ΔCLflap	Combined CLmax
0.0° (Clean)	0.0000	1.4700
10.0° (Takeoff)	+0.2000	1.6700
20.0° (Approach)	+0.3000	1.7700
30.0° (Landing)	+0.3500	1.8200

3. **Elevator Lift Increment**:

- JSBSim Property: `aero/coefficient/CLDe`
- Formula: `ΔCL = 0.4300 · δe`

4. **Alpha-Rate Unsteady Lift Lag**:
   - JSBSim Property: `aero/coefficient/CLadot`
   - Formula: `ΔCL = 1.7000 · [ (α̇ · c̄) / (2 · Vt) ]`
5. **Pitch-Rate Lift Damping**:
   - JSBSim Property: `aero/coefficient/CLq`
   - Formula: `ΔCL = 3.9000 · [ (q · c̄) / (2 · Vt) ]`

---

7. Axis: SIDE (Body Y-Axis Force)

Mathematical Symbolic Equation:

```
Y = q̄ · Sw · CY

Where the total sideforce coefficient CY is decomposed as:
CY = CYβ(β, δf) + CYδa · δa + CYδr · δr + CYp(α, δf) · p̂ + CYr(α, δf) · r̂
```

Stability & Control Derivatives:

```
Weathercock Sideforce Gradient: CYβ  = ∂CY / ∂β   = -0.3926 per rad  (-0.00685 /
deg) (Clean)
Aileron Sideforce Coupling:     CYδa = ∂CY / ∂δa  =  0.0000
Rudder Sideforce Power:         CYδr = ∂CY / ∂δr  = +0.1870 per rad  (+0.00326 /
deg)
```

Components & Tables:

1. **Sideforce Due to Sideslip (CYβ)**:
   - JSBSim Property: `aero/coefficient/CYb`
   - Table Data:

β (rad)	β (deg)	Flaps 0°	Flaps 30°	Physical Effect
-0.3490	-20.0°	+0.1370	+0.1060	Restoring force pushing right
0.0000	0.0°	0.0000	0.0000	Equilibrium
+0.3490	+20.0°	-0.1370	-0.1060	Restoring force pushing left

2. **Rudder Sideforce Control Power**:
   - JSBSim Property: `aero/coefficient/CYdr`
   - Formula: `ΔCY = 0.1870 · δr`

3. **Roll-Rate Induced Sideforce (CYp)**:
    - JSBSim Property: `aero/coefficient/CYp`
    - Table Data:

α (rad)	α (deg)	Flaps 0°	Flaps 30°
0.0000	0.00°	-0.0750	-0.1610
0.0940	5.39°	-0.1450	-0.2310

4. **Yaw-Rate Induced Sideforce (CYr)**:
    - JSBSim Property: `aero/coefficient/CYr`
    - Table Data:

α (rad)	α (deg)	Flaps 0°	Flaps 30°
0.0000	0.00°	+0.2140	+0.1620
0.0940	5.39°	+0.2670	+0.2150

---

8. Axis: ROLL (Body X-Axis Moment)

Mathematical Symbolic Equation:

```
𝒟 = q̄ · Sw · bw · Cl

Where the total rolling moment coefficient Cl is decomposed as:
Cl = Clβ(β) + Clp · p̂ + Clr(α, δf) · r̂ + Clδa · δa + Clδr · δr
```

Stability & Control Derivatives:

```
Dihedral Effect (Roll-Due-to-Beta): Clβ  = ∂Cl / ∂β   = -0.09226 per rad
(-0.00161 / deg)
Roll Damping Derivative:            Clp  = ∂Cl / ∂p̂   = -0.4840
Yaw-Rate Induced Roll:              Clr  = ∂Cl / ∂r̂   = +0.0798 to +0.2317
Aileron Roll Control Power:         Clδa = ∂Cl / ∂δa  = +0.2290 per rad  (+0.00400
/ deg)
Rudder-Induced Roll Coupling:       Clδr = ∂Cl / ∂δr  = +0.0147 per rad  (+0.00026
/ deg)
```

Components & Tables:

1. **Dihedral Effect / Roll Due to Sideslip (Clβ)**:
    - JSBSim Property: `aero/coefficient/Clb`
    - Table Data:

β (rad)	β (deg)	Cl Value	Physical Significance

β (rad)	β (deg)	Cl Value	Physical Significance
-0.3490	-20.0°	+0.0322	Left sideslip induces right restoring roll
0.0000	0.0°	0.0000	Neutral
+0.3490	+20.0°	-0.0322	Right sideslip induces left restoring roll

- **Stability Proof**: `Clβ < 0` guarantees positive dihedral lateral stability.

2. **Roll Damping Derivative (Clp)**:
   - JSBSim Property: `aero/coefficient/Clp`
   - Formula: `ΔCl = -0.4840 · [ (p · bw) / (2 · Vt) ]`
   - **Spin Mechanics Note**: In attached flight, `Clp = -0.4840` provides strong roll damping. In deep post-stall (α > 16°), flow separation reverses `Clp > 0`, turning damping into **roll autostirring**, sustaining autorotation during spins!

3. **Roll Due to Yaw Rate (Clr)**:
   - JSBSim Property: `aero/coefficient/Clr`
   - Table Data:

α (rad)	α (deg)	Flaps 0°	Flaps 30°
0.0000	0.00°	+0.0798	+0.1246
0.0940	5.39°	+0.1869	+0.2317

   - **Physical Note**: Advancing wing in yaw has higher velocity, producing higher lift and rolling the plane into the turn.

4. **Aileron Roll Control Power (Clδa)**:
   - JSBSim Property: `aero/coefficient/ClDa`
   - Formula: `ΔCl = 0.2290 · δa`

5. **Rudder-Induced Roll Coupling (Clδr)**:
   - JSBSim Property: `aero/coefficient/Cldr`
   - Formula: `ΔCl = 0.0147 · δr`

---

9. Axis: PITCH (Body Y-Axis Moment)

Mathematical Symbolic Equation:

```
𝓜 = q̄ · Sw · c̄ · Cmaero + q̄ind · Sw · c̄ · Cmpropwash

Where:
Cmaero     = Cm0 + Cmα · α + Cmq · q̂ + Cmα̇ · α̂̇ + ΔCmflap(δf)
Cmpropwash = Cmδe · δe
```

Stability & Control Derivatives:

```
Zero-Alpha Pitching Moment:        Cm0   = +0.1000
Longitudinal Static Stability:     Cmα   = ∂Cm / ∂α   = -1.8000 per rad  (-0.0314
/ deg)
Pitch Damping Derivative:          Cmq   = ∂Cm / ∂q̂   = -12.4000
Alpha-Rate Downwash Lag:           Cmα̇ = ∂Cm / ∂α̂ = -7.2700
Elevator Pitch Control Power:      Cmδe  = ∂Cm / ∂δe  = -1.1220 per rad  (-0.01958
/ deg)
```

Components & Tables:

1. **Zero-Alpha Pitch Trim Moment (Cm0)**:
    - JSBSim Property: `aero/coefficient/Cmo`
    - Symbol: `Cm0 = +0.1000`
2. **Longitudinal Static Pitch Stability (Cmα)**:
    - JSBSim Property: `aero/coefficient/Cmalpha`
    - Formula: `ΔCm = -1.8000 · α`
    - **Stability Proof**: `Cmα = -1.8000 < 0` satisfies FAA Part 23 positive stick-fixed static longitudinal stability (nose pitches down on upward gust).
3. **Pitch Rate Damping (Cmq)**:
    - JSBSim Property: `aero/coefficient/Cmq`
    - Formula: `ΔCm = -12.4000 · [ (q · c̄) / (2 · Vt) ]`
    - **Dynamics Note**: Provides heavy pitch damping, preventing divergent pitch oscillations and stabilizing the short-period mode.
4. **Alpha-Rate Downwash Lag (Cmα)**:
    - JSBSim Property: `aero/coefficient/Cmadot`
    - Formula: `ΔCm = -7.2700 · [ (α̇ · c̄) / (2 · Vt) ]`
5. **Elevator Pitch Control Power with Propwash Coupling (Cmδe)**:
    - JSBSim Property: `aero/coefficient/Cmde`
    - Formula: `ΔCm = -1.1220 · δe · (q̄ind / q̄)`
    - **Slipstream Coupling**: Scaled by `q̄ind`. When the pilot applies full power during takeoff roll or stall recovery, the propeller wash increases tail dynamic pressure, providing pitch control authority even at very low forward airspeed!
6. **Flap Pitch Trim Change (ΔCmflap)**:
    - JSBSim Property: `aero/coefficient/Cmdf`
    - Table Data:

Flap Position δf	ΔCmflap Increment	Aerodynamic Effect

Flap Position δf	ΔCmflap Increment	Aerodynamic Effect
0.0°	0.0000	Clean baseline
10.0°	-0.0654	Mild pitch-down trim change
20.0°	-0.0981	Moderate pitch-down trim change
30.0°	-0.1140	Strong pitch-down trim change

---

10. Axis: YAW (Body Z-Axis Moment)

Mathematical Symbolic Equation:

```
𝒩 = q̄ · Sw · bw · Cnaero + q̄ind · Sw · bw · Cnpropwash

Where:
Cnaero     = Cnβ(β) + Cnp · p̂ + Cnr · r̂ + Cnda · δa
Cnpropwash = Cnδr · δr
```

Stability & Control Derivatives:

```
Directional / Weathercock Stability: Cnβ  = ∂Cn / ∂β   = +0.05874 per rad
(+0.001025 / deg)
Roll-Rate Induced Adverse Yaw:       Cnp  = ∂Cn / ∂p̂   = -0.0278
Yaw Damping Derivative:              Cnr  = ∂Cn / ∂r̂   = -0.0937
Adverse Aileron Yaw Coupling:        Cnda = ∂Cn / ∂δa  = -0.0053 per rad
(-0.000092 / deg)
Rudder Directional Control Power:    Cnδr = ∂Cn / ∂δr  = -0.0430 per rad
(-0.00075 / deg)
```

Components & Tables:

1. **Directional Static Stability / Weathercock (Cnβ)**:
   - JSBSim Property: `aero/coefficient/Cnb`
   - Table Data:

β (rad)	β (deg)	Cn Value	Physical Meaning
-0.3490	-20.0°	-0.0205	Left sideslip induces left restoring yaw
0.0000	0.0°	0.0000	Directional equilibrium
+0.3490	+20.0°	+0.0205	Right sideslip induces right restoring yaw

   - **Stability Proof**: `Cnβ = +0.05874 > 0` guarantees positive weathercock static directional stability.

2. **Roll-Rate Induced Yaw Coupling (Cnp)**:

- JSBSim Property: `aero/coefficient/Cnp`
- Formula: `ΔCn = -0.0278 · [ (p · bw) / (2 · Vt) ]`

3. **Yaw Damping Derivative (Cnr)**:

- JSBSim Property: `aero/coefficient/Cnr`
- Formula: `ΔCn = -0.0937 · [ (r · bw) / (2 · Vt) ]`
- **Dynamics Note**: Resists yaw rotation, dampening the Dutch roll oscillation mode.

4. **Adverse Aileron Yaw (Cnda)**:

- JSBSim Property: `aero/coefficient/Cnda`
- Formula: `ΔCn = -0.0053 · δa`
- **Flight Mechanics Note**: Downgoing aileron creates higher induced drag than upgoing aileron, yawing the nose opposite to the commanded roll.

5. **Rudder Directional Control Power with Propwash Coupling (Cnδr)**:

- JSBSim Property: `aero/coefficient/Cndr`
- Formula: `ΔCn = -0.0430 · δr · (q̄ind / q̄)`
- **Slipstream Coupling**: Scaled by `q̄ind`, allowing the pilot to counter P-factor yaw during high-power runway takeoff ground roll.

---

11. 6-DOF Coordinate Frame Transformations in JSBSim C++

In `FGAerodynamics.cpp`, forces calculated in wind axes (`LIFT`, `DRAG`, `SIDE`) are transformed into the aircraft body axes, and moments are transferred from the Aerodynamic Reference Point (AeroRP) to the Center of Gravity (CG):

1. Wind-to-Body Coordinate Transformation Matrix (Tw2b):

```
        [  cos(α)·cos(β)   -cos(α)·sin(β)   -sin(α) ]
Tw2b =  [        sin(β)           cos(β)         0   ]
        [  sin(α)·cos(β)   -sin(α)·sin(β)    cos(α) ]
```

2. Body Force Transformation:

```
Fbody_x = - Drag · cos(α)·cos(β) + Side · sin(β)       + Lift · sin(α)·cos(β)
Fbody_y =   Drag · cos(α)·sin(β) + Side · cos(β)       - Lift · sin(α)·sin(β)
Fbody_z = - Drag · sin(α)                              - Lift · cos(α)
```

3. Transfer of Moments from AeroRP to Center of Gravity:

```
MCG = MAeroRP + (rCG - rAeroRP) × Fbody
```

```
Where:
Δx = xCG - xRP
Δy = yCG - yRP
Δz = zCG - zRP

Roll_CG  (LCG) = Roll_RP  + (Δy · Fbody_z - Δz · Fbody_y)
Pitch_CG (MCG) = Pitch_RP + (Δz · Fbody_x - Δx · Fbody_z)
Yaw_CG   (NCG) = Yaw_RP   + (Δx · Fbody_y - Δy · Fbody_x)
```

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12. Master Aerodynamic Summary Formula Matrix

```
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|                                    CESSNA 172P COMPLETE AERODYNAMIC FORMULATION
MATRIX                               |
+─────────────────┬────────────────────────────────────────────────────────────────────────────────────
──────────────┬──────────────────────+
| FORCE / MOMENT | MATHEMATICAL GOVERNING SYMBOLIC EQUATION
| PRIMARY DERIVATIVE |
+─────────────────┼────────────────────────────────────────────────────────────────────────────────────
──────────────┼──────────────────────+
| Drag (D)        | D = q̄ · Sw · [ CD0 + kCDge·(ΔCDflap + CDwbh) + CDδe·|δe| +
CDβ·|β| ]          | CD0 = 0.0270        |
| Lift (L)        | L = q̄ · Sw · [ kCLge·(CLwbh + ΔCLflap) + CLδe·δe + CLα̇·α̇ +
CLq·q̂ ]          | CLα = 4.4700/rad    |
| Sideforce (Y)   | Y = q̄ · Sw · [ CYβ + CYδr·δr + CYp·p̂ + CYr·r̂ ]
| CYβ = -0.3926/rad  |
| Roll Mom (ℒ)    | ℒ = q̄ · Sw · bw · [ Clβ + Clp·p̂ + Clr·r̂ + Clδa·δa + Clδr·δr ]
| Clp = -0.4840       |
| Pitch Mom (ℳ)   | ℳ = q̄·Sw·c̄·[ Cm0 + Cmα·α + Cmq·q̂ + Cmα̇·α̇ + ΔCmflap ] +
Cmδe·δe             | Cmα = -1.8000/rad   |
| Yaw Mom (𝒩)     | 𝒩 = q̄·Sw·bw·[ Cnβ + Cnp·p̂ + Cnr·r̂ + Cnda·δa ] + Cnδr·δr
| Cnβ = +0.0587/rad  |
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```